MIL-M-49275(CR)

26 February 1987

MILITARY SPECIFICATION

MASTS, QUICK ERECT,EXPANDABLE AB-1309(V)/TRC

GENERAL SPECIFICATION FOR

This specification is approved for use within the Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for performance, design, development, and testing for the 112 foot (34 meter) Mast AB-1309/TRC.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

W-R-550	Rod, Ground (With Attachments)
QQ-A-200	Aluminum Alloy, Bar, Rod, Shapes, Tube, and Wire, Extruded, and Structural Shapes, General Specifications for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronics Command, ATTN: AMSEL-ED-TO, Fort Monmouth, New Jersey, 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5985

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QQ-A-225	Aluminum & Aluminum Alloy Bar, Rod, Wire, or Special Shapes, Rolled, Drawn, or Cold Finished, General Specifications for
QQ-A-250	Aluminum and Aluminum Alloy Plate and Sheet, General Specifications for
GGG-H-86	Hammer, Hand, (Forged Steel Head)

MILITARY

MIL-P-116	Preservation, Methods of
MIL-M-809	Mitten Insert, Trigger Finger, O.g. 208
MIL-M-834	Mitten Set, Extreme Cold Weather
MIL-G-835	Glove Inserts, Wool Nylon, M-1949
MIL-A-3777	Anchors, Guy and Rods, Anchor
MIL-A-3962	Anchor, Ground, Arrowhead with Driving Equipment
MIL-W-6858	Welding, Resistance, Spot and Seam
MIL-M-8090	Mobility, Towed Aerospace Ground Equipment, General Requirements for
MIL-P-11268	Part, Materials, and Processes Used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-I-23053/5	Insulation Sleeving, Electrical, Heat Shrinkable, Polyolefin, Flexible Crosslinked
MIL-W-45205	Welding, Gas Metal-arc and Gas Tungsten-arc, Aluminum Alloys, Readily Weldable for Structures, Excluding Armor
MIL-C-46168	Coating, Alphatic Polyurethane, Chemical Agent Resistant
MIL-H-46855	Human, Engineering Requirements for Military Systems, Equipments and Facilities
MIL-P-47158	Penetrant Inspection, Soundness Requirements for Materials, Parts and Weldments
MIL-S-81506	Steel Plate, Sheet, and Strip, 17-4 Ph, Corrosion Resistant, Precipitation Hardening

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STANDARDS

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-209	Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Equipment
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-1261	Welding Procedures for Constructional Steels
MIL-STD-1472	Human Engineering Design Criteria For Military Systems, Equipment and Facilities

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

US Army Electronics Command

SC-C-539482 Extinguisher (Hand - 5 LB)

(Copies of specifications, standards, and other Governments documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

Advisory Circular AC 70/7460-1F	Department of Transportation Federal Administration of Aviation
ANSI Z35.1-1972	American National Standards Institute 1430 Broadway, New York, NY 10018
Article 410-56(d)	National Electrical Code National Fire Protection Association 470 Atlantic Ave., Boston, MA 02210

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ASTM-A-582 ASTM-D-2000	American Society for Testing Materials 1916 Race St. Philadelphia, PA 19103
Installation Requirements for Lightning Protection Systems	Underwriters Laboratories, Inc. 333 Pfingsten Road Northbrook, IL 60062

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3.0 REQUIREMENTS

3.1 Specification Sheets The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.1.1 Description - The AB-1309/TRC Mast system consists of a trailer, telescoping mast sections, base support and guy system, coaxial cable transmission line deployment and recovery capability systems, two each 10KW diesel power generators, control panel, aircraft warning lights, lightning protection, and all the necessary hardware, tools and facilities for storage of antennas and other mission items. The mast and its accessories shall be capable of being erected and retracted from and to the transportation mode by personnel wearing arctic clothing.

3.1.2 First Article.

When specified in the contract or purchase order the contractor shall furnish First Article units in accordance with 4.3.1.

3.2 Characteristics.

3.2.1 Performance of the Mast System. The mast system shall meet the following requirements:

3.2.1.1 Two Stages. The mast shall be deployable in two stages. The lower stage shall have four moving sections and the upper stage shall have two moving sections.

3.2.1.2 Slope. The mast system shall have provisions to operate on terrain that is sloped 10 degrees except when facing down the slope it is 5 degrees.

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3.2.1.3 Deployment. The mast system shall be deployable in winds up to 25 MPH in any direction. (4.10.1 thru 4.10.12)

3.2.1.4 Pointing Accuracy. Each antenna shall not be deflected more than plus or minus 1.5 degrees (cone angle of 3.0 degrees max) from its initial position by winds up to 60 MPH in any direction when the mast is fully deployed or at any intermediate height. (4.10.13 thru 4.10.18)

3.2.1.5 Survival. The mast system, at any height, shall survive an 80 MPH wind in any direction, have no permanent deformation, be redeployable, and meet the pointing accuracy requirement 3.2.1.4 (4.10.19 thru 4.10.24)

3.2.1.6 Deployment Time. Complete deployment of the mast system starting from the transportation mode in winds shall not exceed 25 MPH, on slopes shall not exceed 10 degrees, using the antenna configuration of 3.2.2.4, and a three man crew shall be accomplished in 60 minutes maximum. (4.11)

3.2.1.7 Recovery Time. Using the conditions of 3.2.1.6 the fully erected mast system shall be returned to the transportation mode in 60 minutes maximum. (4.12)

3.2.1.8 Functional Test. The mast system shall use the antenna configuration of 3.2.2.4 or as designated by another specification or contract. The trailer may be on level ground with the wind not exceeding 25 MPH. Permanent anchors may be used. The mast system shall be fully deployed and all electrical, mechanical, and manual features or systems functionally tested. (4.9)

3.2.2 Physical Characteristics.

3.2.2.1 Trailer. The mast system shall be trailer mounted.

3.2.2.1.1 Weight. The weight of the mast system in the heaviest configuration shall not exceed 15,000 pounds gross (including all system and anciliary items). (4.13)

3.2.2.1.2 Mast System Envelope. The mast system shall not exceed 29 feet 2 inches + 2 inches length, 8 feet + 0/-1 inches width, and 98 inches +0/-1 inches in height. (4.14)

3.2.2.1.3 Construction. The construction shall be of light weight material to the maximum extent possible. No welding shall be used on the main load carrying members. The running gear design shall be equivalent to that of a 5 ton modified M979 suspension as made by Southwest Mobile Systems Corp., St. Louis, Missouri.

3.2.2.1.4 Trailer Forward Wheel. The trailer forward wheel shall be of a pnuematic type to facilitate positioning of the trailer in off the road situations.

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3.2.2.1.5 Storage. The trailer shall be equipped with storage compartments for anchors, guys, tools, antennas, and other items associated with the mast system. In addition, storage shall be provided for items designated in subsequent specifications or in the contract.

3.2.2.1.6 Generators. The trailer shall mount two MEP-003A 10 Kilowatt, 60 cycle diesel engine generators.

3.2.2.1.7 Fire Extinguisher. Two fire extinguishers shall be provided. They shall be CO2, Type 1, Size 5 per Drawing SC-C-539482. They shall be mounted in readily accessible locations.

3.2.2.1.8 Electrical circuits. All electrical circuits shall employ circuit breakers in accordance with MIL-STD-252.

3.2.2.1.9 Center of Gravity. The center of gravity of the total mast system for each configuration shall be located such that the lunette load shall be 200 to 500 pounds with the mast in the transportation mode. Also the load on the side wheels shall be within 5 percent of each other based on the total mast system weight. The coordinates of the Center of Gravity (for each configuration) shall appear on a nameplate. (4.16)

3.2.2.1.10 Leveling. The mast shall be capable of being plumbed, vertical, when the trailer is positioned in any direction on terrain that is sloped 10 degrees maximum except when it is facing down the slope it shall be 5 degrees maximum. This may be accomplished by the combination use of the mast tilting mechanism and the leveling jacks. Three manual leveling jacks, two on the rear of the trailer and one on the front, shall be provided. The jacks shall have bearing plates capable of supporting the mast system weight where the contact with the earth does not exceed the requirements of 3.2.2.2.8. The jacks shall retract flush with the lower chassis line of the trailer, ideally, but shall not protrude below this plane by more than 2 inches. A level bubble shall be provided with a calibration ring at zero and 10 degrees. (4.17)

3.2.2.1.11 Towing. The mast system shall be capable of being towed by a 5 ton M-923 truck with full safety lighting and braking interface. (4.18)

3.2.2.1.12 Control Panel. A control panel shall be provided that is readily accessible from ground level. It shall contain all the controls, instruments, gages, etc, necessary for the operation of the mast system. Covers or doors shall be provided for protection against the environments and road hazards. The azimuth and elevation indicators and controls and the wind velocity indicator display and alarm shall be in a single module. The modular items shall be functional when mounted in the control panel on the trailer or in the shelter. A 50 foot extension cable shall be provided for remote module operation. (4.15)

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3.2.2.1.13 Spare Tire. A spare tire mounted on a wheel shall be furnished. The assembly shall be identical to the running gear used on the trailer. The spare tire mount shall be designed to permit two man operation and not require disassembly of any other portion of the trailer. (4.57)

3.2.2.2 Mast. The mast (both stages) shall be telescopic and made of light weight material. Each mast section structural components shall be constructed such that assembly is achieved by bolting or riveting to the largest extent possible. If welding is used it shall be heat-treated to its original condition. Limit switches shall be provided for extension and retraction. A fail-safe feature (to both stages) shall be provided to protect the system if the extension limit switches fail. A manual back-up system shall be provided for the lower stage. (4.19)

3.2.2.2.1 Tilting Mechanism. The time required to tilt the mast from the horizontal to the vertical position shall not exceed two minutes and shall be accomplished by means of two single stage hydraulic actuators. A manual back-up system shall be provided. The pistons shall be provided with dust covers. The hydraulic actuators shall be fully extended when the mast is vertical; that is, perpendicular to the trailer when the trailer is leveled. (4.20)

3.2.2.2.2 Mast Platform. The top of the uppermost mast section shall be designed to be common for mounting different Mast Platforms required for different antenna configurations. The antenna Az-El Actuators (Az-El means Azimuth-Elevation) shall mount to the Mast Platform. For the AS-1425 Antenna the spacing(s) shall be about seven feet.

3.2.2.2.3 Interlock Switches. (4.21)

a. An interlock switch shall be provided to prevent the operation of the mast tilting mechanism if the mast tie down is not undone or removed.

b. An interlock switch shall be provided to prevent the operation of the mast until the tilt mechanism has raised the mast to the vertical position.

c. An interlock switch shall be provided to prevent the tilt mechanism from actuation when the mast has been extended.

3.2.2.2.4 Mast Height. The mast height shall be 114 feet +6/-0 inches minimum as measured from the ground surface (level) to the top surface of the uppermost mast section (where the mast platform mounts). (4.22)

3.2.2.2.5 Mast Extension Time. The time required to extend the mast to its full height after being placed in the vertical position shall not exceed six minutes. (4.23)

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3.2.2.2.6 Mast Section Locks. Mast section locks shall be incorporated to achieve positive locking between the lower two mast sections of the lower stage. (4.24)

3.2.2.2.7 Mast Brakes. Electric brake systems shall be incorporated into the input shafts of the lift mechanisms (both stages) which locks the input shafts when power to the lift motors is switched off. (4.25)

3.2.2.2.8 Pads, Support. The pads alone that support the mast system facing up a 10 degree slope shall be of sufficient area not to exceed a earth contact pressure of 3000 PSF under a 80 MPH wind in any direction. (4.26)

3.2.2.2.9 Height Indicator. A tower Height Indicator shall be used. It shall have a digital readout with an accuracy of plus or minus 1.0 foot, does not lose its memory when the electric power is shut off, and is located in the non-modular portion of the control panel. (4.27)

3.2.2.2.10 Wind Velocity Indicator. An anemometer shall be placed at the top of the mast with a wind velocity range up to 80 miles per hour. Its accuracy shall be plus or minus 2 miles per hour. It shall have a feature that will trigger an alarm when a preset velocity is reached. The display and alarm shall be mounted in the modular portion of the control panel. (4.28)

3.2.2.3 Guying. Provisions for guying the mast shall be included.

3.2.2.3.1 Anchors. The following anchors shall be provided:

a. Six each 6 inch diameter screw anchors; Type IV, Class B per MIL-A-3777.

b. Twelve each, Size 8 inch, Grade B, style 4 anchors per MIL-A-3962 (with retrieval cable and driving equipment).

3.2.2.3.2 Anchor Driving Tools. The following anchor driving tools shall be provided:

a. Hammer, Engineers per Federal Specification GGG-H-86.

b. PIONJAR Rock Drill Breaker System with necessary attachments to deploy anchors. The paint finish shall be per the mast system. (4.29)

3.2.2.3.3 Guy Reels. Individual guy wire storage reels shall be provided. A stand shall be provided to support the reels during payout and hand cranking for retrieval. Means to prevent free-wheeling of the reels shall be provided. Means to secure the stowed guy shall be provided. (4.30)

3.2.2.3.4 Radius Rope. A Radius Rope shall be provided for the location of the guy wire anchors. Attachment points shall be located or designated on the trailer that

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are to be used with the Radius Rope. (4.32)

3.2.2.4 Antennas. The mast system shall use three AS-1425 Antennas and three low loss coaxial cables. Provisions for two AS-1425 Antennas/low loss coaxial cables and two or three AS-3047 Antennas/coaxial cables shall be provided when required by contract or other specifications.

3.2.2.4.1 Antenna Adapter(s). Suitable adapters, if necessary, shall be provided to interface with the Antenna Az-El Actuators.

3.2.2.4.2 Antenna Az-El Actuator. Each Antenna Az-El Actuator shall be capable of providing remotely controlled azimuth orientation of plus or minus 110 degrees from a midpoint position and from the horizonal minimum elevation angles of plus 90 degrees above to 18 degrees below when connected to its respective transmission cable. The azimuth speed shall be variable, reversible from zero RPM to 2.0 RPM.

The elevation control shall be reversible and have a speed of 0.1 $\pm$.01 RPM. (4.33)

a. The Actuators shall not have more than 0.05 degrees of backlash.

b. Limit switches shall be placed at the ends of travel.

c. The Actuators shall have sufficient strength in bending and torsion under all wind conditions.

3.2.2.4.3 Az/El Indicators. One Azimuth Indicator and one Elevation Indicator shall be provided to serve all the (three) antennas. It shall have an adjustable zero (true north) orientation feature, analog or digital readout, and be accurate to plus or minus 1 degree. (4.34)

3.2.2.4.4 Payload. For payload purposes the mast system shall use the antenna configuration of 3.2.2.4.

3.2.2.5 Stall Load. The following items shall be designed so that they are capable of stalling their driving force(s) without mechanical or electrical failure. (4.35)

a. Mast Tilt, both ends.

b. Each Antenna Control, azimuth, both ends.

c. Each Antenna Control, elevation, both ends.

3.2.3 Reliability.

3.2.3.1 Mast Cycling. The mast shall be cycled continuously from the stowed position to the fully erected position 250 times. There shall be no adjustments,

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failures, or replacement of parts in any 25 cycle increment. A minimum of maintenance and adjustment at 25 cycle increments is permitted but a major overhaul is considered to be a failure. Replacement of minor parts are permitted after 125 cycles. (4.36)

3.2.3.2 Azimuth Cycling. Each antenna shall be cycled its full excursion 200 times at 2.0 ±.1 RPM and 200 times at 0.1±.01 RPM. There shall be no failures. (4.37)

3.2.3.3 Elevation Cycling. Each antenna shall be cycled its full excursion 400 times with no failures at a rate of 0.1±.01 RPM. (4.38)

3.2.4 Maintainability. Provisions for maintenance shall be designed into the mast system. The use of special hardware or tools to accomplish maintenance shall be avoided.

3.2.5 Environmental Conditions. The configuration per 3.2.2.4 or as designated by other specifications or contract shall be subjected to the following environmental conditions and shall be met without degradation of performance:

3.2.5.1 Temperature. (4.39)

a. Non-operating and/or storage. The mast system shall be capable of non-operating and/or storage at temperatures between -80 degrees F to +160 degrees F.

b. Operating. The equipment shall be capable of initial and/or continuous operation in temperatures ranging from -50 degrees F to +125 degrees F. To the high temperature condition shall be added solar radiation heat build up of 360 $BTU/FT^2/HR$.

3.2.5.2 Sand and Dust. The mast system shall withstand and be capable of continuous operation when exposed to dry sand and dust. (4.40)

3.2.5.3 Rain. The mast system shall withstand and be capable of continuous operation when exposed to rain. (4.41)

3.2.5.4 Humidity. The mast system shall withstand and be capable of continuous operation when exposed to humidity. (4.42)

3.2.5.5 Salt Fog. The mast system shall withstand and be capable of continuous operation when exposed to salt fog. (4.43)

3.2.5.6 Sunshine. The mast system shall withstand and be capable of continuous operation when exposed to sunshine consisting of visible light, ultra-violet and infrared radiation. (4.44)

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3.2.5.7 Fungus. The mast system shall utilize materials that are fungus resistant as described in MIL-STD-454, Requirement 4. (4.45)

3.2.5.8 Ice Load. The mast system when coated with a layer of ice 0.5±.1 inches thick shall be capable of being: (4.46)

a. Deployed from the transportation mode.

b. Returned to the transportation mode from the deployed mode.

3.2.6 Transportation.

3.2.6.1 Lift and Tiedown Points. Lift and tie down points shall be provided in accordance with MIL-STD-209. (4.47)

3.2.6.2 Fixed Wing Aircraft. The mast system shall be capable of being transported by the following types of aircraft: C-130, C-141, and C-5A without disassembly. (4.31)

3.2.6.3 Helicopter. The mast system shall be capable of being transported by helicopter. (4.48)

3.2.6.4 Rail-Impact. The mast system shall be capable of being transported by rail. It shall withstand Rail Impact. (4.49)

3.2.6.5 Ship. The mast system shall be capable of transportation by inland, coastal, and sea vessels. (4.31)

3.2.6.6 Roadability Tests. (Type V mobility, See 4.5.12 of MIL-M-8090F). (4.50)

Surface/Test Course	Maximum Speed (MPH)	Average Speed (MPH)	Distance (Miles)
Highways - paved roads (Perryman straightaway)	60	50	600
Gravel roads (Munson gravel road course)	20	10	100
Cross-country (level)	20	10	200
Belgian block (Munson belgian block)	20	10	100

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3.2.6.7 Fording Ability. The equipment shall be designed for fording to the depth of three feet. The equipment shall remain in the water for at least 15 minutes. There shall be no failure or impending failure to any of the equipment. (4.51)

3.3 Design and Construction.

3.3.1 Materials, Processes, and Parts. Shall be per MIL-P-11268.

3.3.1.1 Materials. The materials for all components shall be as follows:

a. All aluminum parts shall be made of QQ-A-200, QQ-A-225, or QQ-A-250.

b. All exposed gears shall be corrosion resistant steel (CRES) 17-4PH condition H925 per MIL-S-81506.

c. All pins, shafts, pulleys, miscellaneous hardware, etc shall be CRES Type 303 per ASTM-A-582.

d. All cables shall be aircraft stainless steel, flexible, and preformed.

e. Any rubber parts shall conform to ASTM-D-2000.

3.3.1.2 Processes.

a. Finish: The mast system shall use finishes in accordance with MIL-F-14072. All painted surfaces shall be properly prepared in accordance with Specification MIL-C-46168 with final coat of forest green, Chemical Agent Resistant Coating (CARC) paint.

b. Welding. When welding is required it shall conform to MIL-W-45205 for aluminum alloys, MIL-STD-1261 for metal arc welding of steel, and MIL-W-6858 for spot, stitch, and seam welding.

3.3.1.3 Parts

a. The sheave or drum diameter to cable diameter shall not be less than 24 to 1. (4.52)

b. The fleeting angle of any cable shall be less than 3.0 degrees. (4.53)

c. All cables(s)/end fittings(s) joint(s) shall meet the rated breaking (tension) load of the cable without failure. The test shall be with a suitable length of cable. (4.54)

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d. All cables (assembly) shall fail at or above their rated loads before their end fittings fail (loosen or slip). (4.55)

e. Tools. The use of tools required for the operation of the mast system shall be held to a minimum but are to include three tensiometers. Tools shall be supplied for on-site maintenance.

3.3.1.4 Controls and Tools. The controls and tools shall be designed and/or supplied to be functional under arctic conditions. The controls and tools shall be operable by the crew in arctic clothing where the hands are covered by Mitten Set, Extreme Cold Weather per MIL-M-834 (shell and liner) and Glove Inserts, Wool Nylon, M-1949, per MIL-G-835 or Mitten Insert, Trigger Finger, O.g.208, per MIL-M-809. (4.56)

3.3.2 DELETED.

3.3.3 Nameplate and Product Markings. All items, parts, units, assemblies, etc shall be suitably marked for identification. The marking shall conform to MIL-M-13231.

3.3.3.1 Each basic trailer and mast assembly shall have its own nameplate and be consecutively serialized at starting at number 1.

3.3.3.2 Each configuration shall have its own nameplate and be consecutively serialized starting at number 1.

3.3.4 Workmanship. The mast system shall be manufactured and assemblied in accordance with Requirement 9 of MIL-STD-454. (4.58)

3.3.5 Interchangeability. All parts shall be physically and functionally interchangeable without modification of the parts. Individual parts shall not be hand picked for fit performance. Reliance shall not be placed on any unspecified dimension, rating, characteristics, etc.

3.3.6 Safety engineering. (4.59) The equipment design criteria shall be in accordance with Title 29, Code of Federal Regulations, Chapter XVII, Part 1910, "Occupational Safety and Health Standards", Requirement 1, MIL-STD-454; and paragraphs entitled "Labeling" and Hazards and Safety" of MIL-STD-1472, and shall include but not be limited to the following:

3.3.6.1 Personnel safety. The system shall be designed to minimize the probability and severity of injury to personnel throughout the systems life cycle. Personnel safety shall not be considered a candidate for trade-off.

3.3.6.1.1 Mechanical safety. The system design shall include the safety provisions called for in paragraph entitled "Mechanical Hazards of Requirements 1, MIL-STD-454

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and the following:

a. Exposed edges and corners shall be rounded sufficiently to minimize laceration/puncture hazards, the design goal being a minimum radius of 0.04 inch (1 mm) for edges and a minimum of 0.5 inch (12.7 mm) for corners.

b. Adequate provisions for lifting and handling shall be provided in accordance with paragraph entitled "Unit design for Efficient Handling", MIL-STD-1472.

c. Adequate safeguards shall be installed to prevent inadvertent entrapment of body parts and clothing in moving parts.

d. Color coding of indicator lights shall conform with Table II, MIL-STD-1472.

3.3.6.1.2 Electrical safety. The system design shall include the safety provisions called for in paragraph entitled "Electrical" of Requirement 1, MIL-STD-454 and the following:

a. Accidental contact with current-carrying metal components, terminals, and like devices (between 70 and 500 volts), if such points are exposed to contact during direct support or operator maintenance, shall be controlled by the following:

(1) Wire connections (bare terminals) shall be protected with insulated sleeving (MIL-I-23053/5).

(2) All contacts; terminals, and like devices other than wire connections shall be provided with barriers or guards to prevent operating and maintenance personnel from accidental contact. These barriers or guards shall be manufactured from a transparent, nonconductive material (insulation resistance shall be not less than 100 megohms) and shall be marked with the approximate highest voltage (nearest round number) and current type which may be encountered upon removal. The marking shall be in a format in accordance with American National Standards Institute (ANSI) Z35.1-1972 and worded as follows: "CAUTION (Maximum Voltage applicable) Volts AC". (or DC when applicable).

(3) Small openings shall be provided (where needed) in guards or barriers for maintenance testing without the removal of the guards. The voltage encountered through these openings shall be marked on the guard or barrier.

b. Components over 500 volts shall be completely enclosed and interlocked with no bypass devices permitted and shall be marked in accordance with paragraphs entitled "MARKING" and "SIGNS" Requirement 1, MIL-STD-454. NOTE: Modularized or sealed high voltage assemblies which are opened only at depot may be exempt from interlocking requirements when approved by the procuring activity.

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c. Positive acting, automatically actuated discharging devices shall be provided to discharge high voltage circuits and capacitors to 30 volts within 2 seconds or less. (See 3.3.6.1.3 g below)

d. Attachment plugs shall have dead-front construction per 1981 NEC Article 410-56(d).

3.3.6.1.3 Grounding. The system design shall comply with the following provisions to assure all noncurrent-carrying metal parts are grounded by noncurrent-current conductors:

a. Requirement 1, MIL-STD-454, Paragraphs entitled as follows: "Grounding Potential,""Grounding," "Grounding to Chassis," "Shielding," and "Shorting rods,".

b. All power input or interconnecting cables shall be provided with a green insulated equipment ground wire (EGW) used only for grounding noncurrent-carrying metal parts of equipment. Test equipment leads shall have provisions to insure that the chassis of equipment under test is also at ground potential.

c. The equipment ground wire shall be terminated at both ends in the same manner as the other conductors and shall be of such size and current rating as to conduct safely any fault currents that may be imposed upon it. It shall carry no current at any time other than during a ground fault.

d. All grounding terminals shall be green.

e. All power return wires (neutral) shall be white and white shall be used only for power return wires (neutrals).

f. The equipment ground wire shall not be connected to the grounded power return (neutral) at any point other than the generator or the secondary of the power distribution transformer. (Example: SEPARATION OF NEUTRAL FROM SHELTER FRAME GROUND).

g. To allow for isolation between neutral and ground, and where bleeder resistors are utilized to discharge filter capacitors, connections are to be made between phases and neutral (not ground). No bleeders are needed on filters inserted in neutral line as neutral and ground are tied together at source.

h. In all electrical and electronic equipment, the AC power neutral (white wire) shall be isolated from the equipment, ground by a minimum of one (1) megohm DC resistance measured at the power input connector.

i. Color coding for all power cables and power distribution wiring to rack level shall comply with the following:

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(1) Single phase a.c. service (such as 115 volt):

Black (hot)
White (neutral)
Green (equipment ground wire)

NOTE: Ground is essential; 3-wire cable is required.

(2) Three phase a.c. circuits:

Black (hot) Phase A
Red (hot) Phase B
Blue (hot) Phase C
White (neutral)
Green (equipment ground wire)

NOTE: Ground is essential; 5-wire cable is required.

(3) Single phase, tapped a.c. (such as 110/220):

Black (hot)
Red (hot)
White (neutral)
Green (equipment ground)

NOTE: Ground wire is essential; 4-wire cable required.

j. All outside convenience outlets shall be wired to incorporate ground fault circuit interrupters (GFCI).

3.3.6.2 Aircraft Warning Lights. Two or more Aircraft Warning Lights, red, shall be affixed to the Mast Platform. The lights shall rise higher than the antenna dishes, not interfere with them, nor be blocked or screened by them. The lights shall be capable of being turned on, off, or made to blink. The design shall be guided by the Advisory Circular AC 70/7460-1F, Dept of Transportation, Federal Aviation Administration. (4.60)

3.3.6.3 Lightning Protection. The mast system shall be protected from lightning. The system shall contain an air terminal, down conductor and ground rod. The materials shall be compatible with the mast system materials. The down conductor shall be continuous, have 100 ampere minimum rating, bend radii shall be greater than 8 inches, and bonded to the air terminal. The ground rod shall be a 9 foot sectional Type III, Class B per W-R-550. A clamp shall interface between the down conductor and ground rod with a feature for slack removal due to variations of installations. See standard for Installation Requirements for Lightning Protection Systems, UL96A, Ninth Edition, Underwriters Laboratories, Inc. (4.61)

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3.3.7 Human Engineering. Human Engineering design criteria included in MIL-STD-1472 shall be applied so as to maximize the military effectiveness of man-equipment combinations and minimize unnecessary demands upon skill, training, and manpower. This equipment shall be accomplished by the utilization of the human engineering requirements established in MIL-H-46855. Data shall be supplied in accordance with requirements established in MIL-H-46855. (4.62)

3.4 Precedence. Conflicts of documentation or requirements shall be resolved by submission to the Contracting Officer.

4 QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Test equipment and inspection facilities. The manufacturer shall certify that test and inspection facilities of sufficient accuracy, quality and quantity are established and maintained to permit performance of required inspections.

4.2 Classification of Inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection. (See 4.3.2) Does not include packaging.

b. Inspections covered by subsidiary documents. (See 4.4).

c. Quality conformance inspections.

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(1) Quality conformance inspection of equipment before packaging. (See 4.5).

(2) Packaging inspection. (See 4.63).

4.3 First Article Test (FAT) Program. First Article inspections shall be performed in accordance with the approved test plans and test procedures. Satisfactory completion of the FAT program is a mandatory requirement for final approval of the first article. The order of contractor performed inspections, unless otherwise specified, shall be in accordance with the approved first article test plan. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor. The first article inspection shall consist of inspections specified in Table II. All test plans and procedures shall be submitted for government review and approval prior to testing.

4.3.1 First Article Units. The contractor shall furnish two (2) first article units of the nomenclatures as defined by the contract.

4.3.2 First Article Inspection. The first article inspections shall consist of the inspections specified in subsidiary documents covering the items listed in Table I and the inspections specified for Group A, Group B, Group C and Group D (see Tables III, IV, V, and VI, respectively). The inspection shall be performed in the following order: para 4.4, Group A and Group B for all units, and designated units for specific subgroups within Group C in accordance with Table II, and Group D.

4.4. Inspection Covered by Subsidiary Documents. The mast system shall be inspected under the applicable subsidiary documents as given in Table I as part of the inspection required by this specification, and the inspection requirement specified in the contract.

4.5 Quality Conformance Inspection of Equipment Before Packaging. The contractor shall perform the inspections specified in para 4.4 and Tables III, IV, V, and VI in that order. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A Inspection. Each mast system required to be delivered on contract shall be inspected for conformance with the inspection requirements specified in Table III. Discrete lots shall be formed from items that pass this 100% inspection. Factors of lot composition not defined herein, or in the contract, shall be subject to sampling inspection utilizing the procedures of MIL-STD-105, the special inspection level S-4, and a 1% AQL for major defects and 4% AQL for minor defects.

4.5.1.1. Order of Inspection Within Group A. Group A inspection shall be performed in the order shown in the Government approved First Article Test Plan.

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4.5.2 Group B Inspection. Group B inspection shall be performed on items that have passed Group A 100% and sampling inspection. This inspection shall conform to Table IV. Group B inspection shall be performed in the order shown in the Government approved First Article Test Plan. The procedures for small sample inspection of MIL-STD-105 using the special inspection level S-4 shall apply unless otherwise specified.

4.5.3 Group C Inspection. Group C Inspection shall be performed on units that have passed Group A and Group B Inspection. The inspections shall consist of the inspections specified in Table V. Samples shall be selected in accordance with para 4.5.3.1. The initial Group C inspection on contract shall be the Group C inspection performed on FAT samples.

4.5.3.1 Sampling for Group C Inspection.

4.5.3.1.1 Sub-Group I. For this sub-group, one unit shall be randomly selected from the first production lot. For subsequent Group C inspection, one unit shall be selected at random from each five (5) units or fraction thereof produced during the remainder of the contract.

4.5.3.1.2 Sub-Group II. For this sub-group, one unit shall be randomly selected from the first production lot. For subsequent Group C inspection, one unit shall be selected at random from each twenty (20) units or fraction thereof produced during the remainder of the contract.

4.5.3.1.3 Sub-Group III. For this sub-group, one unit shall be randomly selected from the first production lot. For subsequent Group C inspection, one unit shall be selected at random from each fifty (50) units or fraction thereof produced during the remainder of the contract.

4.5.3.1.4 Sub-Group IV. For this sub-group, one unit shall be randomly selected from the first production lot. This shall cover Group C requirements for the first 500 units. For subsequent Group C inspection, one unit shall be selected at random from each 500 units or fraction thereof produced during the remainder of the contract.

4.5.3.2 Order of Inspection Within Group C. Group C inspection shall be performed in accordance with the order cited in MIL-STD-810 for environmental and durability tests, and in accordance with the order shown in the approved First Article Test Plan for other tests not included in MIL-STD-810.

4.5.3.3 Reinspection of Conforming Group C Sample Units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

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4.5.4 Group D Inspection, Production Reliability Acceptance Test (PRAT). Group D inspection shall consist of reliability inspection in Table VI, and shall be performed on units which passed Group A and Group B inspections. The first Production Reliability Acceptance Test (PRAT) on contract shall be the PRAT performed on FAT samples. Unconditional approval of the First Article by the Contracting Officer will serve as the basis for the verification of compliance with the specified reliability requirements for equipment items delivered during the period up to completion of the next Group D inspection specified herein to be performed on contract. Subsequent Group D inspections for the remainder of the contract shall be performed on one test item randomly selected from every fifty (50) units or fraction thereof from conforming lots that have passed Groups A and B inspections.

4.5.4.1 Reinspection of Conforming Group D Sample Equipment Items. Unless otherwise specified, sample units which have been subjected to and passed Group D inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.6 Test Scheduling Responsibilities. The contractor is responsible for assuring that the schedule developed for the performance of inspections specified herin will enable and assure availability of the equipment items required for performance of these inspections in order to insure timely completion and compliance with the contract delivery schedule.

4.7 DELETED.

4.8 DELETED.

4.9 DELETED.

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TABLE I Subsidiary Documents

Item	Reg. Para
Construction (running gear)	3.2.2.1.3
Fire Extinguisher	3.2.2.1.7
Electrical Circuits	3.2.2.1.8
Mast (welding/heat-treating)	3.2.2.2
Anchors (screw)	3.2.2.3.1.a
Anchors (triangle)	3.2.2.3.1.b
Anchors Driving Tool (Hammer)	3.2.2.3.2.a
Anchors Driving Tool (source inspection)	3.2.2.3.2.b
Mat'l, Processes & Parts	3.3.1
Materials (aluminum)	3.3.1.1.a
Materials (CRES, 17-4PH)	3.3.1.1.b
Materials (CRES, Type 303)	3.3.1.1.c
Materials (CRES, Cable)	3.3.1.1.d
Materials (rubber)	3.3.1.1.e
Process (finish)	3.3.1.2.a
Process (welding)	3.3.1.2.b
Nameplate and Prod. Marking	3.3.3
Workmanship	3.3.4

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TABLE II First Article Inspection

Inspection	Req. Para	Test Para	Sample No. 1	Sample No. 2
Inspections Covered by Subsidiary Documents	SEE TABLE I		X	X
Group A Inspection	SEE TABLE III		X	X
Group B Inspection	SEE TABLE IV		X	X
Group C Inspection	SEE TABLE V			
Subgroup I			X	
Subgroup II				X
Subgroup III				X
Subgroup IV			X	
Group D Inspection	SEE TABLE VI			X
Safety	3.3.6	4.59	X	
Human Engineering	3.3.7	4.62		X

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TABLE III Group A Inspection

Inspection	Req. Para	Test Para	Accept-Reject Criteria
Visual & Mechanical			No Failures
Two Stages	3.2.1.1	4.64	Allowed
Construction (no weld main members)	3.2.2.1.3		"
Trailer Forward Wheel (pneumatic)	3.2.2.1.4	4.64	"
Storage (all items)	3.2.2.1.5	4.64	"
Generators (provisions for)	3.3.2.1.6	4.64	"
Leveling (pad flushness)	3.2.2.1.10	4.64	"
Control Panel (access, cover/doors)	3.2.2.1.12	4.64	"
Tilting Mechanism (hyd. leak, piston covers)	3.2.2.2.1	4.64	"
Mast Platform (bolted on, antenna spacing)	3.2.2.2.2	4.64	"
Radius Rope (attachment I.D. Points)	3.2.2.3.4	4.64	"
Antenna Adaptor(s)	3.2.2.4.1	4.64	"
Maintainability	3.2.4	4.64	"
Design and Construction	3.3	4.64	"
Tools (quantities)	3.3.1.3.e	4.64	
Functional			
Leveling (trailer bubble)	3.2.2.1.10	4.17.1	NO FAILURES
Leveling (jack function)	3.2.2.1.10	4.17.2	ALLOWED
Control Panel (module 2 plcs)	3.2.2.1.12	4.15	"
Mast (back-up function)	3.2.2.2	4.19.d	"
Tilt Mechanism (spirit level)	3.2.2.2.1	4.20	"
Tilt Mechanism (limit switches)	3.2.2.2.1	4.20	"
Tilt Mechanism (back-up functions)	3.2.2.2.1	4.20	"
Interlock switches	3.2.2.2.3	4.21	"
Mast Extension Locks (function)	3.2.2.2.6	4.24	"
Mast Brakes (function, no load)	3.2.2.2.7	4.25	"
Height Indicator (function)	3.2.2.2.9	4.27	"
Wind Vel Ind (function/alarm)	3.2.2.2.10	4.28	"
Guy Reels	3.2.2.3.3	4.30	"
Az-El Actuator (function)	3.2.2.4.2	4.33.c	"
Az/El Indicators (function)	3.2.2.4.3	4.34	"
Aircraft Warning Lights	3.3.6.2	4.60	"

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TABLE IV Group B Inspection

Inspection	Req. Para	Test Para	AQL
Interchangeability	3.3.5	4.66	4%
Mast System Envelope	3.2.2.1.2	4.14	4%
Radius Rope (lengths)	3.2.2.3.4	4.32	4%
Parts (cable/drum diameter)	3.3.1.3.a	4.52	4%
Parts (fleeting angle)	3.3.1.3.b	4.53	4%
Parts (joint tests)	3.3.1.3.c	4.54	4%
Parts (cable assembly)	3.3.1.3.d	4.55	4%
Lightning Protection	3.3.6.3	4.61	4%

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TABLE V Group C Inspection

Inspection	Req. Para	Test Para	Accept-Reject Criteria
Subgroup I			
Mast (Fail-Safe)	3.2.2.2	4.19.a	No Failures
Wind Vel Ind (cal/alarm)	3.2.2.2.10	4.28	Allowed
Subgroup II			
Mast (retraction)	3.2.2.2	4.19.b	No Failures
Tilting Mechanism (timing)	3.2.2.2.1	4.20	Allowed
Mast Extension Time	3.2.2.2.5	4.23	"
Height Indicator (calibration)	3.2.2.2.9	4.27	"
Antenna Az-El Actuator	3.2.2.4.2	4.33	"
Az/El Indicators (calibration)	3.2.2.4.3	4.34	"
Temperature, Non-Operating and/or Storage	3.2.5.1.a	4.39	"
Temperature, Operating	3.2.5.1.b	4.39	"
Subgroup III			
Weight	3.2.2.1.1	4.13	No Failures
Center of Gravity	3.2.2.1.9	4.16	Allowed
Mast Height	3.2.2.2.4	4.22	"
Mast Extension Locks (test)	3.2.2.2.6	4.24	"
Mast Brakes (with load)	3.2.2.2.7	4.25	"
Humidity	3.2.5.4	4.42	"
Ice Load (from xport mode)	3.2.5.8.a	4.46	"
Ice Load (to xport mode)	3.2.5.8.b	4.46	"
Lift and Tie Down Pts	3.2.6.1	4.47	"
Helicopter	3.2.6.3	4.48	"
Fording	3.2.6.7	4.51	"

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TABLE V Group C Inspection (cont'd)

Inspection	Req. Para	Test Para	Accept-Reject Criteria
Subgroup IV			
Deployment	3.2.1.3	4.10.1-4.10.12	No Failures
Point Accuracy	3.2.1.4	4.10.13-4.10.18	Allowed
Survival	3.2.1.5	4.10.19-4.10.24	"
Deployment Time	3.2.1.6	4.11	"
Recovery Time	3.2.1.7	4.12	"
Leveling (jacks, full tipping)	3.2.2.1.10	4.17.2	"
Spare Tire	3.2.2.1.13	4.57	"
Mast (back-up)	3.2.2.2	4.19.c	"
Tilting Mechanism (back-up-full)	3.2.2.2.1	4.20	"
Pads, Support	3.2.2.2.8	4.26	"
Anchor Dr Tls (drive stake)	3.2.2.3.2.b	4.29	"
Radius Rope (stake location)	3.2.2.3.4	4.32	"
Stall Load	3.2.2.5	4.35	"
Controls & Tools (functional)	3.3.1.4	4.56	"
Sand and Dust	3.2.5.2	4.40	"
Rain	3.2.5.3	4.41	"
Salt Fog	3.2.5.5	4.43	"
Sunshine	3.2.5.6	4.44	"
Fungus	3.2.5.7	4.45	"
Towing	3.2.2.1.11	4.18	"
Fixed Wing Aircraft	3.2.6.2	4.31	"
Rail Impact	3.2.6.4	4.49	"
Ship	3.2.6.5	4.31	"
Roadability	3.2.6.6.	4.50	"

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TABLE VI Group D (Reliability) Inspection

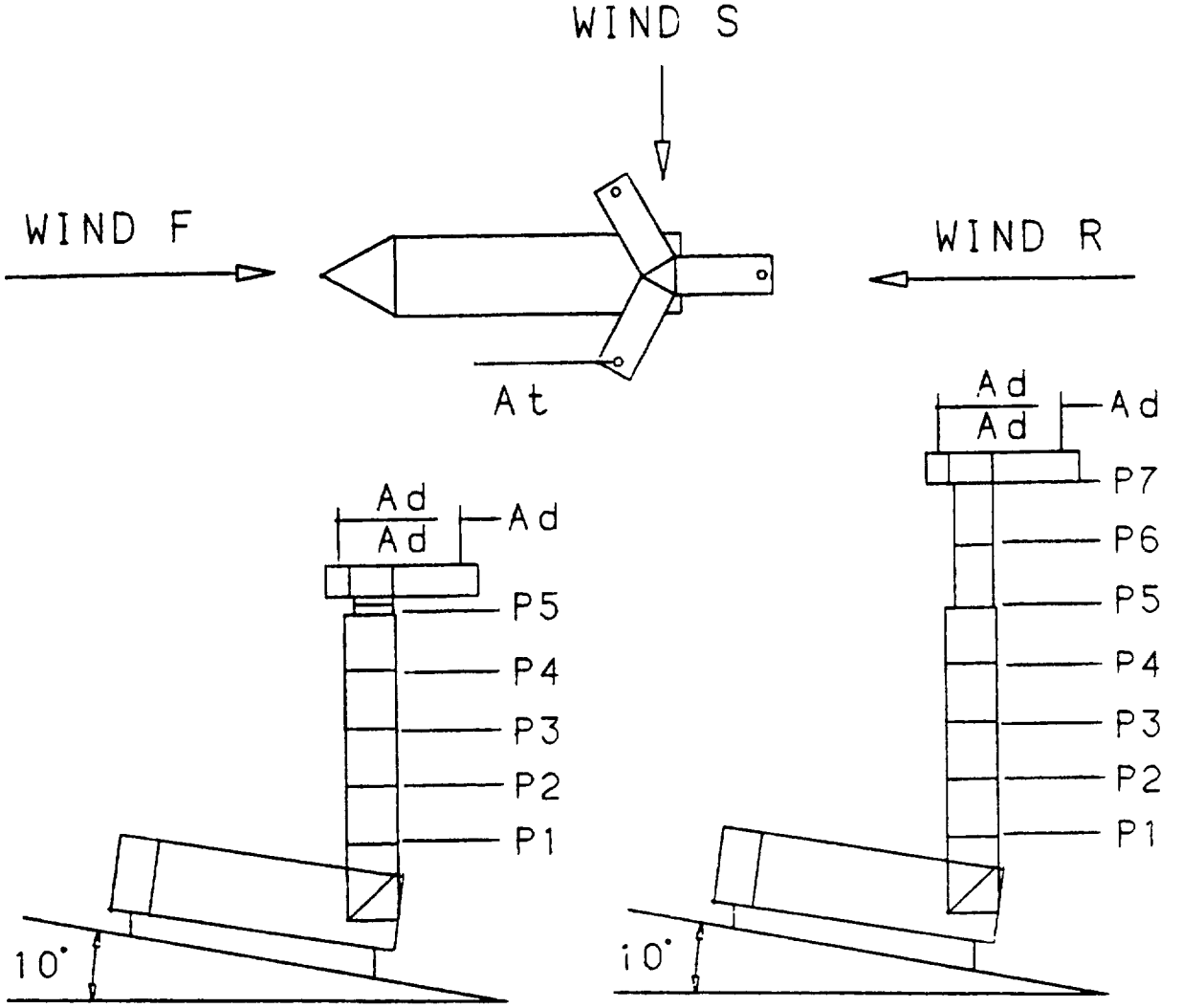
Inspection	Req. Para	Test Para	Accept Reject Critria
Mast Cycling	3.2.3.1	4.36	No Failures Allowed
Azimuth Cycling	3.2.3.2	4.36	"
Elevation Cycling	3.2.3.3	4.36	"

4.10 Wind Load Tests. The requirements of 3.2.1.3, 3.2.1.4, and 3.2.1.5 shall be tested and met under the following conditions: Permanent anchors may be used. The trailer shall be on and facing up a 10 ±.5 degree slope. All loads shall be horizontal. The Pn loads shall be applied to the top of each mast section. The Ad loads shall be applied to the center of pressure of each antenna. The At load shall be applied at the center of pressure to one of the "outboard" antenna post. The Pn loads and the Ad loads or the At load are applied simultaneously. (see FIGURE 1)

TABLE VII. Wind Loads

Pn	Wind Velocities (MPH)			
Loads (lbs)	25 Lower Stage	25 Upper Stage	60	80
P1	115	115	675	1130
P2	100	100	585	945
P3	105	105	615	945
P4	110	100	585	905
P5	85	100	565	840
P6	-	85	480	710
P7	-	50	295	430
Ad	70	75	430	620
At	135	145	835	1210

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FIRST STAGE DEPLOYED

FIRST AND SECOND STAGE DEPLOYED

FIGURE 1
DIRECT WIND LOADS

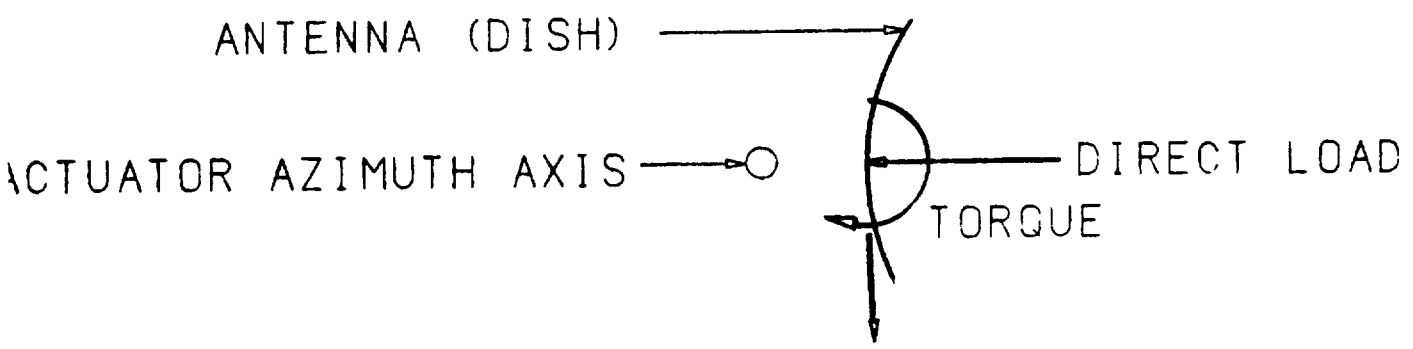


FIGURE 2
ANTENNA WIND LOADS

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4.10.1 The first stage of the mast shall be deployed. Only the guys attached to the upper end of the lower mast section 1 shall be used. The Pn and Ad loads of the 25 MPH Lower Stage column of Table VII shall be applied in the Wind R direction. The mast shall be capable of being lowered 5 ft minimum and raised in the loaded condition.

4.10.2 Repeat 4.10.1 except replace the three Ad loads with the one At load.

4.10.3 Repeat 4.10.1 except with the loads in the Wind F direction.

4.10.4 Repeat 4.10.2 except with the loads in the Wind F direction.

4.10.5 Repeat 4.10.1 except with the loads in the Wind S direction.

4.10.6 Repeat 4.10.2 except with the loads in the Wind S direction.

4.10.7 The first and second stages of the mast shall be deployed. Only the guys attached to the upper ends of mast sections 1 and 5 shall be used. The Pn and Ad loads of the 25 MPH Upper Stage column of Table VII shall be applied in the Wind R direction. The second stage shall be capable of being lowered 5 ft minimum and raised in the loaded condition.

4.10.8 Repeat 4.10.7 except replace the three Ad with the one At.

4.10.9 Repeat 4.10.7 except with the loads in the wind F direction.

4.10.10 Repeat 4.10.8 except with the loads in the wind F direction.

4.10.11 Repeat 4.10.7 except with the loads in the wind S direction.

4.10.12 Repeat 4.10.8 except with the loads in the wind S direction.

4.10.13 The first and second stages of the mast shall be deployed. All the guys shall be used. The Pn and Ad loads of the 60 MPH column of Table VII shall be applied in the Wind R direction. The angles of deflections or total angle of deflection of each antenna shall be measured between their initial positions and their loaded positions.

4.10.14 Repeat 4.10.13 except replace the three Ad with the one At. The resultant angular deflection of the two measured conditions shall meet the Pointing Accuracy of 3.2.1.4.

4.10.15 Repeat 4.10.13 except with the loads in the Wind F direction.

4.10.16 Repeat 4.10.14 except with the loads in the Wind F direction.

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4.10.17 Repeat 4.10.13 except with the loads in the Wind S direction.

4.10.18 Repeat 4.10.14 except with the loads in the Wind S direction.

4.10.19 The first and second stages of the mast shall be deployed. All the guys shall be used. The Pn and Ad loads of the 80 MPH column of Table VII shall be applied in the Wind R direction. The loads shall be held constant for 10 minutes. This test is the first part of the following paragraph.

4.10.20 Repeat 4.10.19 except replace the three Ad with the one At. There shall be no failures of any members, joints, hardware, components, parts, items, etc. There shall be no permanent deformation of members. Remove the wind loads. The mast system shall be fully functional by being returned to the transportation mode and then being redeployed.

4.10.21 Repeat 4.10.19 except with the loads in the Wind F direction.

4.10.22 Repeat 4.10.20 except with the loads in the Wind F direction.

4.10.23 Repeat 4.10.19 except with the loads in the Wind S direction.

4.10.24 Repeat 4.10.20 except with the loads in the Wind S direction.

4.11 Deployment Time. The conditions of 3.2.1.6 shall be tested. The PIONJAR emplacement tool and the 8 inch MIL-A-3962 anchors shall be used. All items shall be mounted and/or connected except the transmission lines to the radios. After the timing period all items shall be operational except the RF properties.

4.12 Recovery time. The conditions of 3.2.1.7 shall be tested. All items shall be recovered and properly stored.

4.13 Weight. (3.2.2.1.1) The mast system in its transportation mode with all its own and otherwise specified equipment shall be weighed. The scale accuracy shall be within plus or minus 100 pounds.

4.14 Mast System Envelope. The mast system envelope shall be measured and meet the requirements of 3.2.2.1.2.

4.15 Control Panel. (3.2.2.1.12) All the controls are to be functionally tested first when the modular unit is in the control panel and then when the modular unit is remotely connected.

4.16 Center of Gravity. The requirements of 3.2.2.1.9 shall be tested and met. The mast system, in the transportation mode and free standing, shall have the following:

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a. A load cell placed under each wheel. The load cells shall have an accuracy of plus or minus 1.0 percent.

b. The lunette load shall be measured while the trailer is level and the forward wheel is off the ground. The measuring device shall have an accuracy of plus or minus 1.0 percent.

4.17 Leveling. (3.2.2.1.10)

4.17.1 The level bubbles' zero and 10 degree ring shall be tested for accuracy against a spirit level in each quadrant before mounting.

4.17.2 The leveling jacks shall be deployed and tested such that they can tip the trailer at least 10 degrees in the curbside and roadside directions. The jacks shall be tested for at least 5 ±.5 degrees tilt in the fore and aft direction.

4.17.3 Test the jacks that they retract at least flush with the lower chassis line of the trailer or not protrude below the specified dimension.

4.18 Towing. (3.2.2.1.11) The towing shall be tested as part of the Roadability tests. (3.2.6.6)

4.19 Mast. Test to the requirements of 3.2.2.2. The mast extension and retraction limit switches shall be tested for their function.

a. Overriding the extension switch shall cause the fail-safe feature to stop the extension action. There shall be no electrical or mechanical failures. This applies to both stages.

b. Overriding the retraction switches shall not cause any electrical or mechanical damage or failures. (The cables may go slack but do not allow them to unwind). This applies to both stages.

c. Test the manual back-up system by fully extending and retracting the lower stage.

d. Test the manual back-up system by demonstrating its ability to extend and retract the mast at least one inch.

4.20 Tilting Mechanism. The requirements of 3.2.2.2.1 shall be tested and met when power is used. The tilting, up and down fully, shall be tested using the manually actuated back-up system. The spirit levels on the mast shall be checked against the true vertical orientation of the mast. The bubble level shall be checked to insure it is mounted correctly. There shall be no hydraulic leakage.

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4.21 Interlock Switches. (3.2.2.2.3)

a. The interlock switch shall fail to operate the tilt mechanism while the mast tie down is not undone or removed. It shall operate it when it is undone or removed.

b. The interlock switch shall fail to operate the extension mechanism while the mast is being tilted to the vertical. It shall operate it when the mast has reached the vertical position.

c. The interlock switch shall fail to operate the tilt mechanism once the mast has been extended two inches.

4.22 Mast Height. The requirements of 3.2.2.2.4 shall be tested and met.

4.23 Mast Extension Time. The requirements of 3.2.2.2.5 shall be tested and met.

4.24 Mast Extension Locks. (3.2.2.2.6) The mast system configured per 3.2.2.4 shall be extended fully and the mast extension locks of the lower stage engaged. The extension mechanism shall lower the mast system slightly such that the lift cables go slack. Hold the system for a five minute time period. Raise the system and remove the locks. Lower system about three feet and repeat two more times. Examine locks and bearings. Any evidence of shear or cold flow shall be cause for failure.

4.25 Mast Brake. (3.2.2.2.7) The mast system shall be configured per 3.2.2.4 plus 200 ±5. pounds added at the Mast Platform, 3.2.2.2.2, centrally located over the mast.

a. Erection. The mast brakes shall stop the erection of each stage within 5 seconds after its lift motor is switched off at the 40 ±4 foot level. The input shaft shall not creep (backwards) when observed for one minute. Repeat at the 80 ±6 foot level and the 113 ±1 foot level.

b. Retraction. The Mast Brake shall stop the retraction of each stage within 10 seconds after the lift motor is switched off at the 80 foot level. The input shaft shall not creep (continue) when observed for one minute. Repeat at the 40 foot level and at the one foot level above the fully retracted position.

4.26 Pads Support. (3.2.2.2.8) Load cells shall be placed under the Pads, Support. The mast shall be fully deployed and subjected to the 80 MPH winds in the directions of R, F, and S. (ref 4.10.19, 4.10.21, and 4.10.23) The resultant soil contact pressure shall not exceed 3000 PSF.

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4.27 Height Indicator. The requirements of 3.2.2.2.9 shall be tested and met. Comparisons of the Height Indicator and the actual measured height shall be made at the 40, 80, and 113 foot levels on the way up and on the way down. At each level the power shall be shut off for 15 seconds and then put on. There shall be no loss or change of the Height Indicator memory.

4.28 Wind Velocity Indicator. (3.2.2.2.10) The anemometer shall be tested in winds that are measured by a standard at every 10 MPH up to 80 MPH. The alarm feature shall be set and tested at 30, 50, and 80 MPH for function.

4.29 Anchor Driving tool. (3.2.2.3.2.b) The PIONJAR Rock Drill Breaker System shall demonstrate its insertion ability on the anchors of 3.2.2.3.1.c and any Ground Rods supplied. Inspect paint finish per the mast system.

4.30 Guy Reels. (3.2.2.3.3) Each guy reel shall be tested that it pays out smoothly, stops immediately when the pay out load is removed (does not free wheel), and cranks in without excessive effort. The device used to secure the stowed guy shall be inspected.

4.31 Transportation. The contractor shall submit data, as required, to the contracting officer or material developer in order that transportability approval from the Commander, MTMC may be obtained as required by AR 70-47.

4.32 Radius Rope. (3.2.2.3.4) The trailer shall be placed on a adequately large, level, and flat area in order to perform this test. The Radius Rope shall be used to spot the anchor locations. These locations shall then be measured referenced to the trailer to insure the Radius Rope meets the requirements of stake locations. Inspect the trailer that the Radius Rope attachment points are properly defined.

4.33 Antenna Az-El Actuator. (3.2.2.4.2) Each Actuator shall be tested for the following:

a. The azimuth range (angle), speed of 2.0 RPM.

b. The elevation range (angle), speed of 0.1 RPM.

c. Starting with variable speed control at zero it shall be increased slowly to its full speed value driving the azimuth to its far end and actuating that limit switch. Repeat the process to the opposite end. The action of the control and antenna shall be smooth and even.

d. The elevation limit switches.

e. The backlash of the azimuth and elevation (gear train).

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f. A load of 590 ±5 lbs (horizontal) shall be applied to the center of pressure of the AS-1425 Antenna. Any permanent deformation shall be a failure. Also with this load applied the full azimuth and elevation shall be traversed. (Each separately tested while the other is at "zero degrees"). While testing one the other shall not creep.

g. In the same manner as f above apply horizontal loads thru the apex of the antenna; a torsional load of 235 ±2 ft-lbs, a direct load of 165 ±2 lbs, and a transverse load of 165±2 lbs. All resultant torques about the Actuator azimuth axis shall be additive. (see figure 2)

4.34 Az/El Indicators. The requirements of 3.2.2.4.3 shall be tested and met. The correlation between the mechanical orientation (azimuth) of the antenna and the Indicator shall be one degree when tested every thirty degrees within the azimuth range. The correlation between the mechanical orientation (elevation) of the antenna andd the indicator shall be within one degree when tested at -18.0, 0.0, +18.0, and +90.0 degrees.

4.35 Stall Loads. The requirements of 3.2.2.5 shall be tested and met. The limit switches at each end are to be overridden. The mechanical stops or drive system is to stall the drive motor. Actuation of bypass means or circuit breakers are not considered to be failures.

4.36 Reliability Test (Group D). A sample of one mast system shall be randomly selected from conforming lots that have passed Groups A and B inspections. The reliability test shall be performed in accordance with the Government approved PRAT test plans and test procedures.

4.36.1 Reliability Test Conditions. The PRAT shall be conducted under ambient environmental conditions.

4.36.2 Parameters to be Measured. The parameters to be measured shall be as determined by the contractor and specified in the Government approved PRAT test plan.

4.36.3 Failure Definition. Failure is defined as the inability of the equipment to perform its required function. This includes catastrophic failures as well as any event which causes departure from performance as required by the equipment specification (paras 3.2.3.1, 3.2.3.2, and 3.2.3.3). A relevant failure shall include failures as defined in paragraph 5.6.1 of MIL-STD-781C and any new failure or degradation of government furnished equipment and government loaned equipment caused by deficiencies in contractor furnished equipment. Analysis of failures shall be in accordance with paragraph 5.6.2 of MIL-STD-781C. Determination of compliance shall be in accordance with paragraph 5.7 of MIL-STD-781C.

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4.37 DELETED.

4.38 DELETED.

4.39 Temperature. (3.2.5.1) The mast system storage configuration shall be the transportation mode. NOTE: Care should be taken to use the correct gear box lubricant while conducting the tests of this paragraph.

a. High. Test per MIL1-STD-810D, Method 501.2, Procedure I.

(1) Step 3. The initial test conditions are ambient conditions.

(2) Step 4. The high temperature shall be +160°F. Humidity need not be controlled. Seven 24 hour cycles shall be used.

(3) Step 5. The controlled ambient temperature shall be +125°F.

(4) Step 6. Operation shall consist of deployment and functionally testing all electrical, mechanical and manual features of the mast system and returning to the transportation mode.

b. Low. Test per MIL-STD-810D, Method 502.2, Procedure I.

(1) Step 2. The chamber air temperature shall be -80°F.

(2) Step 3. The stabilized -80°F shall be held for 4 hours minimum.

(3) Step 5. Low-temperature operation is required.

Procedure II

Step 1. The chamber air temperature for low temperature operation is -50°F.

(4) Step 4. Operational check out per Procedure I step 6.

(5) Step 6. Manipulation is not required.

(6) Step 10. Operational check out per Procedure I step 6.

4.40 Sand and Dust. (3.2.5.2) The mast system shall be in the transportation mode. Test per MIL-STD-810D, Method 510.2.

a. Procedure I - Blowing dust

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Step 2. The air velocity shall be 1400-1750 ft/min.

Step 4. The exposure shall be 1.5 hours.

Step 5. The adjusted temperature shall be +160°F.

Step 8. Operation not required. Exposure shall be for 1.5 hours.

Step 9. When mast system has returned to ambient conditions repeat for other three sides from step 1.

Step 11. Operate per Para 4.39.a Step 6.

b. Procedure II - Blowing Sand

Step 1. The operating temperature shall be +125°F.

Step 2. The air velocity shall be 3540-5700 ft/min.

Step 3. The sand concentration shall be 1.1 ± .025 g/m^3 (0.033± 0.0075 g/ft^3).

Step 4. Expose each side for 1.5 hours.

Step 5. Operation during exposure is not required.

Step 7. Operate per Para 4.39.a Step 6.

4.41 Rain. (3.2.5.3) Test per MIL-STD-810D, Method 506.2, Procedure I.

a. Transportation mode.

Step 1. The rainfall rate shall be 5.8 in/hr. Expose curbside first.

Step 2. The wind velocity shall be 40 MPH.

Step 3. Operation not required.

Step 4. Expose the roadside, end, and front.

Step 6. Examine all cavities for water entry such as storage compartments, gear boxes, and closed housings such as the control panel and its instruments, gages, and indicators.

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Step 8. Operation not required.

b. Deployed mode. Tilt to the vertical. Each stage shall be capable of being extended at least one foot. Guy first section as required. Repeat conditions above except as noted below.

Step 3. During the last 10 minutes of exposure extend and retract each stage at least one foot. Cycle each Az-El actuator fully for azimuth and elevation.

Step 8. Remove from the test facility and perform operational check out per 4.39.a Step 6.

4.42 Humidity. (3.2.5.4) Test per MIL-STD-810D, Method 507.2, Procedure III. The mast system shall be in the transportation mode.

Step 7. Operation near the end of the fifth and tenth cycle not required. After the tenth cycle remove from test chamber and immediately perform operational test per 4.39.a Step 6. After operational test perform a visual examination.

4.43 Salt-fog. (3.2.5.5) The mast system shall be in the transportation mode. Test per MIL-STD-810D, Method 509.2, Procedure I.

Step 4. Remove from the test chamber and perform operation test per 4.39.a Step 6.

4.44 Sunshine. (3.2.5.6) Test per MIL-STD-810D, Method 505.2, Procedure I. Ideally this test is to be conducted when the mast assembly is a integral unit; however, it may be conducted in three parts: the mast platform and above, the first and second stages, and the trailer assembly.

Step 2. Expose the curbside of the mast assemble to a continuous surface solar radiation of 355-360 $BTU/FT^2/HR$.

Step 3 Continue exposure until a steady state temperature for one hour has been achieved.

Step 4. Operation checkout per Para 4.39.a Step 6.

Repeat by exposing the mast assembly from overhead.

4.45 Fungus. (3.2.5.7) Test per MIL-STD-810D, Method 508.3. The test shall be conducted in the transportation mode. All doors, covers, panels, etc shall be ajar. After inspection operate per Para 4.39.a Step 6.

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4.46 Ice Load. (3.2.5.8) Reference is made to MIL-STD-810D, Method 521.0. The mast system shall be placed in a chamber in the transportation mode. The temperature shall be reduced to 17-23°F. A fine spray of water shall be admitted until at least 0.5 inches of ice builds up on all surfaces. At this point the spray shall be turned off and all the free water be allowed to drain or freeze. When there is no free water present, the test can commence.

a. With the tools supplied, and the arctic gloves it shall be demonstrated that the mast can be deployed. The antennas shall be fully cycled for Az-El.

b. The conditions shall be repeated except that the mast is in the deployed mode. The antennas shall be cycled fully for Az-El. It shall be demonstrated that the deployed mast can be returned to the transportation mode.

4.47 Lift and Tiedown Points. (3.2.6.1) Each lift and tiedown point shall be tested to the requirements of MIL-STD-209.

4.48 Helicopter. (3.2.6.3) The mast system shall be tested as if it were slung below a helicopter. The total load shall be three times the basic weight where the added weight shall be distributed on the mast system in accordance to the existing mass distribution. There shall be no failures or permanent distortion. Data on the sling (rating and leg length) shall be submitted along with data on a spreader if one is used.

4.49 Rail-Impact. (3.2.6.4) The mast system, fully equipped and prepared for transportation, shall be loaded on a railroad flatcar and secured. The test shall be conducted on a flat stretch of track. A 165,000 pound car traveling at 9 ± 1/4 MPH (or equivalent impact energy conditions approved by the government) shall be impacted against the test flatcar coupled to two other empty cars. These cars shall be stationary with their brakes off. Two impacts into each end of the test flatcar shall be performed. The lashing and blocking holding the mast system shall be tight at the start of each impact run. If the lashing or blocking is torn loose by the impact, the run shall be repeated. A damage survey to the mast system shall be made after each impact. There shall be no permanent deformation.

All details and data on the lashing, blocking, test flatcar, and the two additional flatcars shall be submitted.

4.50 Roadability Tests. The requirements of 3.2.6.6 shall be tested and met. The test shall only be run at Aberdeen Proving Grounds, Maryland or an approved alternate. The lighting and braking systems shall be tested. (3.2.2.1.11).

Before and after each test the mast system shall be examined for any deformation, looseness of parts or hardware, weld integrity. Any welds or

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cracks that are suspect shall undergo Penetrant Inspection per MIL-P-47158(MI).

4.51 Fording. The requirements of 3.2.6.7 shall be tested and met. An impending failure is one where water has entered a sealed area or wetted items that are required to operate only when dry.

4.52 Parts (drum/Cable Diameters). The requirements of 3.3.1.3.a shall be tested and met.

4.53 Parts (Fleeting Angle). The requirements of 3.3.1.3.b shall be tested and met.

4.54 Part (Cable(s)/Fitting(s) Joints). The requirements of 3.3.1.3.c shall be tested and met. The test shall be one out of every fifty joints produced.

4.55 Parts (Cable, Assemblies). One cable per lot of each distinctive size, length, and designed end fitting(s) shall be tested to the requirements of 3.3.1.3.d. A lot shall be a group of 250 units or less.

4.56 Controls and Tools. The requirements of 3.3.1.4 shall be tested and met. Each control and tool supplied shall be demonstrated that it function can be accomplished when the hands are covered by the full mitten set.

4.57 Spare Tire. The requirements of 3.2.2.1.13 shall be met. It shall be demonstrated that the spare and one of the regular wheels are interchangeable.

4.58 Workmanship. The requirements of 3.3.4 shall be met.

4.59 System Safety Engineering Inspection. (3.3.6) A visual inspection shall be performed to determine that all requirements have been incorporated in the equipment. Inability to meet these requirements shall constitute a failure of the test. Use SEL FORM 1183 to submit data.

4.60 Aircraft Warning Lights. The requirements or 3.3.6.2 shall be tested and met.

4.61 Lightning Protection. The requirements of 3.3.6.3 shall be tested and met.

4.62 Human Engineering. The requirements of 3.3.7 shall be met.

4.63 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First Article Inspection of Packaging.

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b. Quality Conformance Inspection of Packaging.

4.63.1 First Article Inspection of Packaging. Unless otherwise specified in the contract, First Article Inspection of Packaging shall be in accordance with the Unit Pack Design Validation Requirements of MIL-P-116.

4.63.2 Quality Conformance Inspection of Packaging.

4.63.2.1 Materials inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.63.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B Quality Conformance Inspection Requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.63.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in Table VIII "PACKING INSPECTION PROVISIONS." Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and Acceptable Quality level of 4.0 percent defective.

TABLE VIII Packing Inspection Provisions.

NO.	CHARACTERISTIC	METHOD OF INSPECTION
101	Intermediate container not as specified	Visual
102	Improper closure of intermediate container	Visual
103	Shipping containers not in accordance with specification	Visual
104	Excessive cube	Visual
105	Improper blocking and bracing	Visual
106	Closure not in accordance with specification	Visual
107	Weight and size exceed container limitations	Weight & Measure
108	Strapping not in accordance with specification, incorrectly applied, omitted	Visual
109	Marking omitted, incorrect, or illegible	Visual

4.64 Visual and Mechanical Inspection. The mast system shall be given a mechanical (to include dimensional measurements) and visual inspection and test to the requirements of Table III and that quality of all materials and

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workmanship is in compliance with the requirements of this specification. These inspections shall be conducted in accordance with the Government approved FAT plans and procedures. The item shall be examined for the defects listed in MIL-STD-252. Compliance with the following requirements of Section 3 shall also be verified by visual and/or mechanical inspections:

3. _____Warranty Marking (if applicable)
3. _____CARC exterior finish identification marking

4.65 Performance Tests, Assemblies, Modules/PCBs, Subassemblies, Maintenance spares/Repair Parts. Assemblies, modules/PCBs, subassemblies, and maintenance spares/repair parts shall be tested in accordance with their respective test specifications/drawings, as applicable, and the Government approved FAT plan.

4.66 Interchangeability.

4.66.1 Inspection for Dimensional Interchangeability. The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirements of 3.3.5. These inspections sahll be performed in accordance with the Government approved FAT plan/procedures. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

d. Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of replaceable items and major units; and location of the mating parts on the plug-in assembly or major unit.

e. Size and form of special threads.

4.66.2 Inspection for Electrical Interchangeability.

4.66.2.1 Replaceable modules, assemblies, and subassemblies and parts of the mast system identified as maintenance parts/items shall be removed from the mast system and tested in accordance with the applicable requirements of 4.65.

4.66.2.2 Subsequent to satisfactory completion of the applicable tests of 4.65, the modules, assemblies, subassemblies and parts shall be re-installed into a different mast system from which they were removed. No alignment, tuning, etc.,

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other that those authorized by the approved maintenance manuals/documentation is permitted.

4.66.2.3 Subsequent to the reinstallation of the items into the mast system performance tests to verify compliance of the mast system with specified operational requirements shall be performed in accordance with the Government approved test plan. Failure of this mast system to satisfy the specified performance test requirements will constitute test failure.

5. PACKAGING

5.1 Packaging Requirements. The requirements for packaging shall be in accordance with Special Packaging Instruction AG00000531.

6. NOTES

6.1 Intended use. The intended use of this Mast System is to provide the means for the quick erection of various antennas and their transmission lines.

6.2 Ordering information.

6.2.1 Documents. Procurement documents should specify the titles, dates and the number of this specification as well as the variation (slash number) required.

6.2.2 Government testing. Mast Systems are subject to retesting by the Government to insure that they continue to pass the first article or the Functional Test inspection. Failure of a Mast System will not affect its acceptance but may indicate future mandatory action.

6.2.3 Data requirements the following data are required by this specification to be furnished to the Government and should be entered on DD Form 1423 if made part of the contract:

First Article inspection results
Quality conformance inspection results
Contractor furnished inspection equipment designs

6.2.4 Level A or B preservation and packing (See Section 5).

6.2.5 When first article inspection rough handling tests are not required.

6.2.6 When first article packaging inspection test reports require Acquisition Activity approval prior to production unit packing.

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6.3 Level B preservation. When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage, and handling.

6.4 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.5 Subject term (key word) listing.

Accuracy, pointing
Anchors
Antennas
Guying
Load, stall
Mast, quick erect, expandable
Rope, radius
Shapes, extruded and structural
Trailer

Preparing Activity
Army-CR

Custodian
Army-CR

Project No. 5985-A501

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2 DOCUMENT TITLE

3a NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b ADDRESS *(Street, City, State, ZIP Code)*	

5 PROBLEM AREAS

a. Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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